Dual-Semantic Graph Convolution Network for Hyperspectral Image Classification With Few Labeled Samples

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***Abstract*— In recent years, superpixel-based graph convolutional networks (GCNs) have drawn increasing attention within the hyperspectral image (HSI) classification community. Due to the high-dimensional property of HSI, establishing a high-quality and accurate initial graph is still a great challenge for the superpixel-based GCN methods. In addition, the lack of high-level semantics within the superpixel-based node features leads to poor classification performance of the model, especially in scenarios with limited labeled samples. To tackle these problems, we propose a novel approach called the dual-semantic graph convolution network (DSGCN) for HSI classification in this article. Specifically, our method employs superpixel segmentation to construct graph nodes with semantic structure information, treating each superpixel in the HSI as a node within the graph. We design a superpixel-level autoencoder that integrates with the initial graph to update the edge weights. With the learnable edge weights, our model can adaptively learn robust spatial semantic (SS) information from HSI. Additionally, we introduce a spectrum-flow (SF) module to extract global spectral semantic variation information. To further enhance the nonlinearity capability of GCN, we replace the traditional linear layer with a novel network layer referred to as Kolmogorov Arnold networks (KANs) during the node representation phase. In addition, we develop a memory-efficient residual spectral attention (MERSA) module that adapts to the full-batch training manner in the convolutional neural network (CNN) branch to supplement fine-grained pixel-level features. Extensive experiments conducted on four benchmark datasets demonstrate that our proposed DSGCN significantly outperforms several state-of-the-art methods, particularly when using a small amount of labeled data.**

***Index Terms*— Attention mechanism, convolutional neural network (CNN), dual-semantic graph convolution network (DSGCN), hyperspectral image (HSI) classification.**

I. Introduction

HUNDREDS of spectral bands contained in hyperspectral image (HSI) offer the potential for significant performance gains in many fields, such as environmental monitoring, precision agriculture, military applications, and so on [1], [2]. Many traditional HSI classification methods, including support vector machine (SVM) [3], random forest (RF) [4], k-nearest neighbor (KNN) [5], and multinomial logistic regression [6] have been proposed in early researches. However, these traditional models always rely on expert experience or handcrafted features. In contrast, convolutional neural network (CNN) has a significant advantage in capturing the spatial associations between pixels and spectral associations between bands. In recent years, many excellent CNN-based methods have been proposed. In [7], recursive autoencoders (RAEs) were introduced to learn the spatial and spectral information from the original data. A deeper model with spectral and spatial residual blocks was developed in [8] to extract more robust and effective features from HSI. Xie et al. [9] proposed a multiscale densely connected convolutional network (MS-DenseNet) to extract multiple scales information for the classification task. In addition, some works [10], [11] combining CNN with attention mechanism arose successively, where the combination is capable of extracting critical information and suppressing useless information. Nevertheless, there is still one challenging problem for the abovementioned methods, i.e., the overfitting issue caused by the scarcity of labeled samples of HSI.

Different from annotating natural images, the label annotation of HSI requires field knowledge, which is more difficult and time-consuming. From the perspective of model learning, there are mainly three strategies for the problem above: few-shot learning (FSL), self-supervised learning (SSL) [12], and transfer learning [13].

FSL aims to mimic the human ability to recognize novel classes when limited labeled samples are available. Specifically, the generalization ability of the FSL models is improved by randomly selecting subsets of classes from the training set as meta-tasks. More recently, some typical metric-based FSL methods have been widely adopted for HSI classification. In [14], a deep residual 3-D CNN was trained for HSI classification with sampled mini-batches to mimic the few-shot task scenario and learn the prototype representations of each class. Liu et al. [15] proposed a Siamese CNN to extract more discriminative features for HSI classification. In [16], a weighted contrastive loss was designed to improve the ability of the proposed spectral–spatial Siamese network (S3Net) in handling difficult samples. Furthermore, Rao et al. [17] presented a spatial–spectral relation network that considers the profound similarity between samples. In recent years, foundation models based on SSL techniques have drawn

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unprecedented attention in the field of HSI classification. Among these approaches, contrastive-based SSL methods and generative-based SSL methods are the most widely applied. These foundation models are trained through designed pretext tasks to learn transferable knowledge from unlabeled data, which reduces the dependence on annotated data. The core idea of contrastive-based SSL methods is to minimize the distance between positive sample pairs and maximize the distance between negative sample pairs. Some representative frameworks such as momentum contrast (MoCo) [18], simple contrastive learning (SimCLR) [19], and their variants demonstrate excellent classification performance. Generative-based SSL methods [20], [21], [22] focus more on generating new data or recovering complete data. For example, Liu et al. [21] proposed masked spectral reconstruction as the pretext task. The unmasked data are input into the designed autoencoder for masked spectral reconstruction and transferable knowledge learning. Hong et al. [22] developed a universal remote sensing (RS) foundation model with a 3-D generative pretrained transformer (GPT) architecture named SpectralGPT, which was pretrained on one million spectral RS images. SpectralGPT demonstrates good performance across four downstream tasks, including single/multilabel scene classification, semantic segmentation, and change detection. Unlike SSL, transfer learning is suitable for situations where there is a large amount of labeled training data in the source domain. For instance, a classic hyperspectral dataset was selected as the source domain dataset in [23], and a conditional adversarial domain adaptation strategy was utilized to overcome domain shift. However, the abovementioned works only conduct convolution on squared image grids while ignoring the topological information of HSI. When homogenous pixels with long-range connections are encountered, it is difficult for these works to achieve satisfactory performance.

Graph convolutional networks (GCNs) [25] have attracted widespread attention recently due to their effectiveness in handling non-Euclidean data. As a typical semisupervised learning method, the core idea of GCN is to establish the intrinsic relationship among the graph nodes and take full advantage of the label information from limited labeled samples and the rich information from unlabeled samples. Therefore, GCN is very powerful in extracting global topological information. In [26], a semisupervised spectral-spatial GCN (S^2GCN) was presented, which makes full use of the spatial information of the current pixel. Bai et al. [27] designed a spatial–spectral attention layer for the proposed graph attention network (GAT), where part of the useful information is selected for enhancement. However, the importance of exploring the intrinsic relationships between hyperspectral bands is undervalued for the existing GCN-based HSI classification methods. As shown in Fig. 1, it is difficult to classify the two categories from the perspective of pixel value. The classification task becomes more challenging when facing the problem of lack of labeled samples. We can observe that the variations between the bands of the two categories are quite different. It is reasonable to consider variation characteristics as a supplement to the original features.

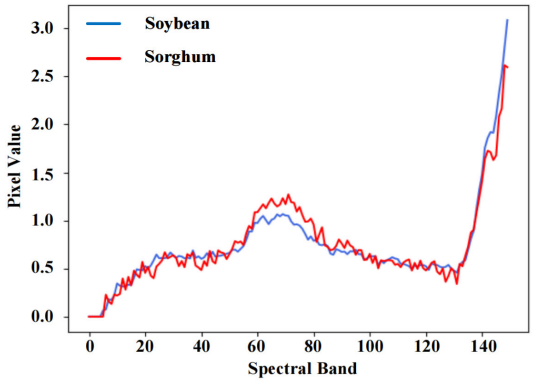


Fig. 1. Spectrum sequence of two categories on the WHU-Hi-HanChuan [24] data (the blue curve represents the soybean category, while the red curve represents the sorghum category).

Based on the aforementioned analysis, this article presents a novel dual-semantic graph convolution network (DSGCN) for HSI classification. To begin, a superpixel-level autoencoder is integrated with the initial graph to obtain robust spatial semantic (SS) information of HSI. The superpixel-level autoencoder performs a similar effect with regularization by adding a reconstruction loss. In this way, the edge weights of the graph are updated when training the model, and the potential abilities of GCN are released. Considering that the inherent relationship and the spectral correlation between hyperspectral bands have not been fully explored in the previous GCN models, we design a spectrum-flow (SF) module to introduce the global spectrum motion feature and enhance the representation of graph nodes for the first time. It is reasonable to enhance GCN with the spectrum motion feature for the following two reasons.

First, GCN has a natural advantage in utilizing global spectral semantic features when training in a full-batch manner. Second, the spectrum motion feature can enrich the feature diversity of graph nodes and improve their representation ability.

In order to further enhance the nonlinear representation and aggregation capabilities of GCN, we replace the linear layer with Kolmogorov Arnold networks (KANs) [28]. The weight parameter of the linear layer becomes the Gaussian radial basis function of KANs.

In addition, we designed a CNN branch with an attention module for exploring pixel-level features. The proposed memory-efficient residual spectral attention (MERSA) module allows the network to focus more on the bands of interest. The features obtained by the GCN and CNN branches are concatenated as the input of a softmax layer for classification.

The main contributions of this article can be summarized as follows.

1) We propose a novel DSGCN for few-shot HSI classification. In the GCN branch, a new GCN component is designed by incorporating a superpixel-level autoencoder with the initial graph. The classification and reconstruction tasks are conducted simultaneously to update the edge weights of the graph. In this way, the

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potential abilities of GCN are released, which is helpful for obtaining robust SS information of HSI.

2) In order to further improve the representation ability of graph nodes, we design an SF module to construct graph nodes with rich global spectral variation features for the first time. Each graph node contains spatial topology information and rich spectral variation information. The KAN layer enhances the nonlinear representation and aggregation capabilities when conducting graph convolution. In addition, we introduce a MERSA module to accommodate the full-batch training manner and extract global contextual features in the CNN branch.
3) We conducted extensive experiments on four well-known HSI datasets to examine the classification performance of the proposed method under extreme conditions (5–25 training samples per class). The experimental results clearly show that the proposed DSGCN outperforms the state-of-the-art algorithms.

The remainder of this article is organized as follows. Section II introduces the relevant progress of GCN for HSI classification. Section III introduces the proposed DSGCN in detail. The experimental results on four public HSI datasets are shown in Section IV. Finally, a conclusion is drawn in Section V.

II. Related Work

In recent years, GCN-based models are getting more attention in the HSI classification community due to the advantages of aggregating neighborhood information and modeling flexibly. Early GCN works attempt to treat each pixel point as a graph node [26]. Mou et al. [29] constructed a nonlocal graph to represent the entire HSI, where each vertex of the graph represents a pixel in the image. However, training the graph that is constructed in this way is computationally excessive. To solve this issue, many works [30], [31], [32], [33] sought to perform graph construction on superpixel-level nodes. First, the original hyperspectral data is segmented into several homogeneous regions by segmentation algorithms like SLIC [34], ERS [35], etc. Pixels with similar spectral signatures and close spatial coordinates are grouped into one superpixel. Then, the adjacency relations between superpixels are calculated to establish the undirected graph. Finally, GCN is performed on the established graph for classification. Zhang et al. [36] proposed a dual-graph autoencoder (DGAE) to fully exploit both the geometric structures of pixels with spatial and spectral information. However, DGAE is implemented in an unsupervised manner, and its performance is not comparable to that of supervised graph methods. Wan et al. [30] proposed a multiscale dynamic graph convolutional network (MDGCN) to generate superpixel-level features, where each superpixel represents a node in the graph. However, MDGCN mainly focuses on the superpixel-level features but ignores pixel-level features, which will lead to information loss. To alleviate this issue, a CNN-enhanced GCN (CEGCN) [31] method was proposed to promote the learning of edge weights. CEGCN can learn features on small-scale regular regions and large-scale irregular regions simultaneously by combining the advantages of CNN and GCN. Following this idea, the attention mechanism was introduced into the GCN and CNN branches in [32], and a weighted fusion of GCN and CNN features was performed in the last layer. Nowadays, similar architectures occupy a significant position in HSI classification [33], [37].

The challenges of the above superpixel-based GCN methods are threefold.

1) The performance of these methods is easily affected by the result of pre-segmentation. Due to the spectral similarity of samples from different classes, one superpixel may contain several categories, resulting in incorrect classification in scenarios with a limited number of labeled samples. For instance, the GCN branch of CEGCN [31] achieves relatively poor results on the Indian Pines dataset due to the influence of pre-segmentation.
2) These models ignore the high-level spatial and spectral semantic information, thereby failing to improve the classification performance further.
3) The robustness of these methods is notably deficient, rendering them inoperative in the presence of noise. In this article, we introduce a novel DSGCN model that takes advantage of the high-level semantic information of HSI. As such, the classification performance and robustness of the GCN model can be improved, especially in the case of small training samples.

III. Proposed Method

In this section, we will describe the proposed DSGCN in detail. As shown in Fig. 2, the proposed network consists of two branches: the GCN branch and the CNN branch. In the GCN branch, there are two important modules designed for exploring semantic information: 1) the SS module and 2) the SF module. In this section, we first describe the SS module and explain how to update the edge weights of the graph by a superpixelwise reconstruction component. Next, the HSI is input into the SF module to extract the global spectral variation information, and the features obtained at each coordinate point are encoded as the node features of the graph. In the CNN branch, the pixel-level features are extracted by the designed MERSA module. Finally, these two types of features are concatenated together to predict the label for each pixel through a softmax classifier.

A. SS Module

Given an HSI, it is represented by $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$, where H denotes its height, W denotes its width, and B denotes the number of bands.

In the first step, the first three principal component analysis (PCA) bands of the raw HSI are input into the SLIC [34] method to generate a perceptually meaningful superpixel set $\{\mathcal{S}_i\}_{i=1}^{N}$, where $\mathcal{S}_i$ represents the ith superpixel in $\mathcal{S}$ and N is the number of superpixels in $\mathcal{S}$. $\mathcal{S}_i$ is comprised of $\{x_j^i\}_{j=1}^{k}$, where x_j^i is the jth pixel in $\mathcal{S}_i$, and k is the number of pixels in $\mathcal{S}_i$. After generating the superpixel map, the HSI can be defined as an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ and $\mathcal{E}$ denote the nodes and edges of $\mathcal{G}$, respectively.

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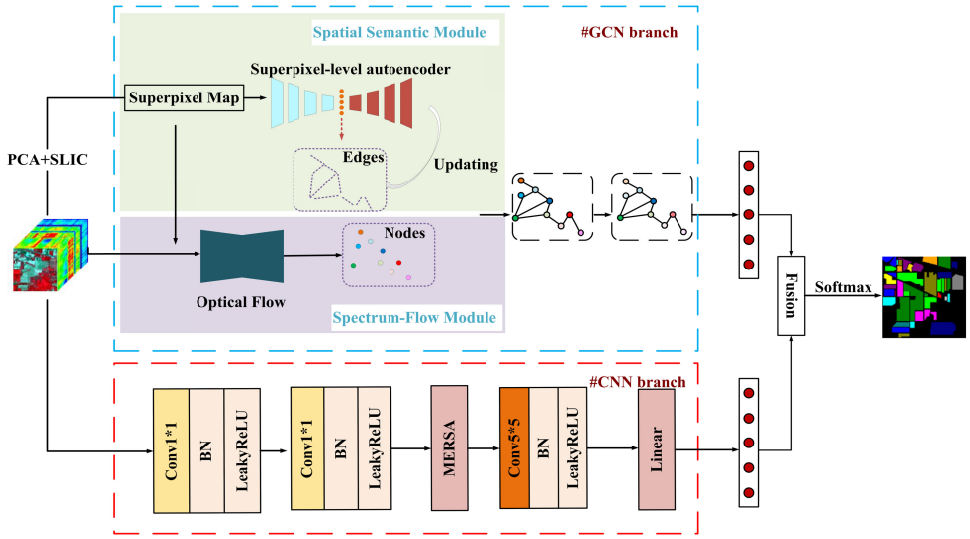


Fig. 2. Schematic illustration of the proposed DSGCN, including the SS module and SF module in the GCN branch and the MERSA module in the CNN branch. The SS module is designed by integrating a superpixel-level autoencoder with the initial graph. The SF module is designed to extract the global spectral variation information. The MERSA module extracts global contextual information within the HSI.

Then, we can establish a coordinate mapping matrix $\mathbf{M} \in \mathbb{R}^{HW \times N}$ between pixels and superpixels

$$\mathbf{M}_{i,j} = \begin{cases} 1, & \text{if } \hat{\mathbf{X}}_i \in \mathcal{S}_j \\ 0, & \text{otherwise} \end{cases} \quad \hat{\mathbf{X}} = \text{Flatten}(\mathbf{X}) \tag{1}$$

where Flatten$(\cdot)$ is a flattening operation for the HSI data by the spatial dimension, and $\hat{\mathbf{X}}_i$ is the ith pixel in $\hat{\mathbf{X}}$. Finally, the matrix form of $\mathcal{V}$ in the graph $\mathcal{G}$ can be obtained with the coordinate mapping matrix, that is,

$$\mathbf{H} = \overline{\mathbf{M}}^{\mathrm{T}} \text{Flatten}(\vartheta(\mathbf{X})) \tag{2}$$

where $\overline{\mathbf{M}}$ denotes the normalized $\mathbf{M}$ by column, and $\vartheta(\cdot)$ denotes the feature extraction function before graph convolution. $\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_N]$. Each node $\mathbf{h}_i$ corresponds to a superpixel $\mathcal{S}_i$.

It is well known that $\mathcal{E}$ is essential in controlling neighbor information aggregation. Particularly, superpixels are very good at conserving image structures, which makes adjacency relations learning more important. For convenience, the edges $\mathcal{E}$ are represented as the adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$. In this section, we propose a new module to learn the adjacency matrix $\mathbf{A}$ by integrating a superpixel-level autoencoder into GCN. As shown in Fig. 2, we first apply PCA to preprocess the HSI for dimensionality reduction. The mean value of the acquired low-dimensional features is used to represent each superpixel for calculating $\mathbf{A}$. We denote the mean feature of each superpixel as v_i. Then, v_i is fed into the designed superpixel-level autoencoder with a Gaussian noise v_{noise}.

The encoding operation is achieved by a series of 1-D convolution and 1-D max pooling with LeakyReLU activation function to obtain latent superpixel-level features. For each node feature v_i, the encoding process is presented as follows:

$$l_i = \phi_e(v_i + v_{\text{noise}}) \tag{3}$$

where ϕ_e denotes the superpixel-level encoder.

TABLE I
NETWORK STRUCTURE OF THE SUPERPIXEL-LEVEL AUTOENCODER

Encoder		Decoder	
Conv1D	1× 1×3, stride=1	ConvTranspose1d	1 × 1 ×3, stride=1
Activation	LeakyReLU	Activation	LeakyReLU
Conv1D	1×1×3, stride=1	ConvTranspose1d	1×1×3, stride=1
Activation	LeakyReLU	Activation	LeakyReLU
MaxPool1d	1, stride=1	ConvTranspose1d	1×1×3, stride=1
Conv1D	1×1×3, stride=1	Activation	LeakyReLU
Activation	LeakyReLU	ConvTranspose1d	1×1×3, stride=1
MaxPool1d	3, stride=1	Activation	LeakyReLU

The corresponding decoding operation adopts 1-D transposed convolution. The decoding process can be simplified as

$$v_i^* = \phi_e^+(l_i) \tag{4}$$

where ϕ_e^+ denotes the decoder and v_i^* is the reconstructed instance of v_i. The architecture of the autoencoder is described in Table I.

On one hand, the latent feature l_i is used to calculate the cosine similarity measurement among nodes. The adjacent matrix $\mathbf{A}$ can be calculated as

$$\mathbf{A}_{i,j} = \begin{cases} \cos(l_i, l_j), & \text{if } \mathcal{S}_i \text{ and } \mathcal{S}_j \text{ are adjacent} \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

where $\cos(\cdot)$ denotes the cosine similarity.

On the other hand, the reconstruction error is added to the total loss function to empower the classification network. Here, we use the Charbonnier loss [38] as the reconstruction loss to avoid the gradient vanishing problem

$$\mathcal{L}_{r1}(v_i, v_i^*) = \sqrt{(v_i - v_i^*)^2 + \varepsilon_1^2}. \tag{6}$$

We empirically set the hyperparameter ε_1 to $1e-6$.

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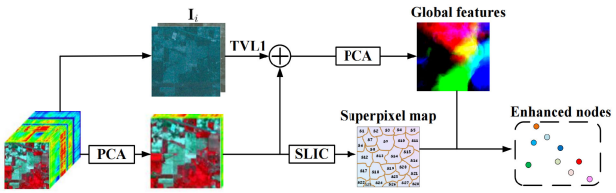


Fig. 3. Schematic of the SF module.

B. SF Module

In view of the data structures of HSI, the critical step for applying superpixel-based GCN to HSI classification is to design reliable and discriminative graph nodes. Therefore, we propose to use the spectral variation features to enhance the representation ability of graph nodes.

It has been proven by Sun et al. [39] and [40] that the motion feature brought by spectral variation can be reflected by the brightness change. For HSI, brightness changes can be analyzed using optical flow estimation methods. Therefore, we designed an SF module based on the TVL1 algorithm [41].

The original HSI can be split into consecutive single-band gray images $\mathbf{B}_i, \mathbf{B}_{i+1} \in \mathbb{R}^{H\times W}(i = 1, 2, \ldots, B-2)$. As shown in Fig. 3, we feed two three-channel images $\mathbf{I}_i = (\mathbf{B}_{i-1}, \mathbf{B}_i, \mathbf{B}_{i+1})$, $\mathbf{I}_{i+1} = (\mathbf{B}_i, \mathbf{B}_{i+1}, \mathbf{B}_{i+2})$ into the TV-L1 algorithm to extract the global spectrum motion feature $\mathbf{F}_m$

$$\mathbf{F}_m = \psi(\mathbf{I}_i, \mathbf{I}_{i+1}) \tag{7}$$

where $\psi(\cdot)$ denotes the TV-L1 algorithm, which can be implemented based on the OpenCV python library. Then, the designed spectral feature $\mathbf{F}$ can be obtained by concatenating $\mathbf{F}_m$ and $\mathbf{X}$, that is,

$$\mathbf{F} = \text{pca}(\text{concat}(\mathbf{F}_m, \mathbf{X})) \tag{8}$$

where pca$(\cdot)$ denotes the PCA for reducing the dimensionality and $\mathbf{F} \in \mathbb{R}^{H\times W\times C'}$. Then, $\mathbf{F}$ is transformed into the initial graph node features $\mathbf{H}^0$ with (2).

In this way, graph nodes are endowed with rich spatial-contextual information and spectral variation information. The classification results of GCN are improved through the designed node features.

C. Graph-Based Feature Extraction

After obtaining the dual-semantic graph nodes, we can define the graph Laplacian matrix as

$$\mathbf{L} = \mathbf{D} - \mathbf{A} \tag{9}$$

where $\mathbf{D}$ is the degree matrix of the graph and its diagonal element is equal to the sum of the weights of all its edges, i.e., $\mathbf{D}_{i,i} = \sum_j \mathbf{A}_{i,j}$. Then the normalized Laplacian matrix is calculated by $\mathbf{L} = \mathbf{D}^{-(1/2)}\mathbf{L}\mathbf{D}^{1/2}$. Finally, the graph's analysis is turned into the Laplace matrix's analysis. Because $\mathbf{L}$ is a symmetric and positive semidefinite matrix, the spectral decomposition of $\mathbf{L}$ can be obtained as

$$\mathbf{L} = \mathbf{I} - \mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}} = \mathbf{U}\Lambda\mathbf{U}^{\mathrm{T}} \tag{10}$$

where $\mathbf{U} = [\mathbf{u}_0, \mathbf{u}_1, \ldots, \mathbf{u}_{n-1}] \in \mathbb{R}^{N\times N}$ and $\mathbf{u}_i$ is the feature vector of the Laplacian matrix $\mathbf{L}$. $\Lambda \in \mathbb{R}^{N\times N}$ is the corresponding matrix that consist of the eigenvalues of $\mathbf{L}$

$$\Lambda = \begin{bmatrix} \lambda_1 & & 0 \\ & & \\ 0 & & \lambda_N \end{bmatrix} \in \mathbb{R}^{N\times N}. \tag{11}$$

The process of convolution in the topological domain of functions f_x and f_y is expressed as

$$f_x *_{\mathcal{G}} f_y = \mathbf{U}\left(\left(\mathbf{U}^{\mathrm{T}} f_x\right) \odot \left(\mathbf{U}^{\mathrm{T}} f_y\right)\right) = \mathbf{U} g_\theta(\Lambda)\mathbf{U}^{\mathrm{T}} f_x \tag{12}$$

where $*_{\mathcal{G}}$ represents the GCN and $\odot$ represents the point multiplication. $g_\theta(\Lambda) = \text{diag}(\theta_1, \theta_2, \ldots, \theta_N)$ denotes the convolutional kernel in the spectral domain. However, due to the high computational complexity, performing GCN in this way is not practical. To circumvent this problem, Hammond et al. [42] introduce the Kth-order Chebyshev polynomials to fit the spectral filter g_θ, that is,

$$g_\beta(\Lambda) \approx \sum_{k=0}^{K} \beta_k \Lambda^k \tag{13}$$

where $\Lambda = \text{diag}(\lambda_1, \lambda_2, \ldots, \lambda_N)$ and β_k is the parameters to be learned. Then, (12) becomes

$$f_x *_{\mathcal{G}} f_y = \mathbf{U} g_\beta(\Lambda)\mathbf{U}^{\mathrm{T}} f_x = \sum_{k=0}^{K} \beta_k \mathbf{L}^k f_x \tag{14}$$

where $\mathbf{L}^k$ denotes the kth power of the Laplacian matrix. By letting $K = 1$ and $\beta' = (1/2)\beta_0 = -\beta_1$, (14) can be further simplified by

$$f_x *_{\mathcal{G}} f_y = \beta_0 f_x + \beta_1 \mathbf{L} f_x = \beta'\left(\mathbf{I} + \mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}\right) f_x. \tag{15}$$

The renormalization trick [43] is introduced to replace $\mathbf{I} + \mathbf{D}^{-(1/2)}\mathbf{A}\mathbf{D}^{-(1/2)}$ by $\widetilde{\mathbf{D}}^{-(1/2)}\widetilde{\mathbf{A}}\widetilde{\mathbf{D}}^{-(1/2)}$, where $\widetilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ and $\widetilde{\mathbf{D}} = \mathbf{D} + \mathbf{I}$. The final graph convolution can be formulated as

$$\mathbf{H}^{l+1} = \Phi\left(\overset{*}{\mathbf{A}}\mathbf{H}W_l\right) \tag{16}$$

where $\overset{*}{\mathbf{A}} = \widetilde{\mathbf{D}}^{-(1/2)}\widetilde{\mathbf{A}}\widetilde{\mathbf{D}}^{-(1/2)}$, Φ denotes the KAN layer, which is designed to perform a nonlinear projection, and W_l is the parameter to be learned.

D. Pixel-Level Convolutional Network

However, superpixel-based GCN methods merely utilize large-scale features and ignore the detailed information of HSI. To address this issue, we adopt CNN to extract pixel-level detail information in this article as a supplement to the GCN branch.

It is well known that inter-band correlations play an important role in interpreting HSIs. However, when training the network in the full-batch manner, the calculation of the inter-band similarity matrix takes $\mathcal{O}(HWB_g^2)$ memory, where HW denotes the total number of pixels in the HSI and B_g denotes the spectral dimension of the input feature map. Taking the Botswana dataset as an example, which has a total number of 377 856 pixels, and assuming the spectral

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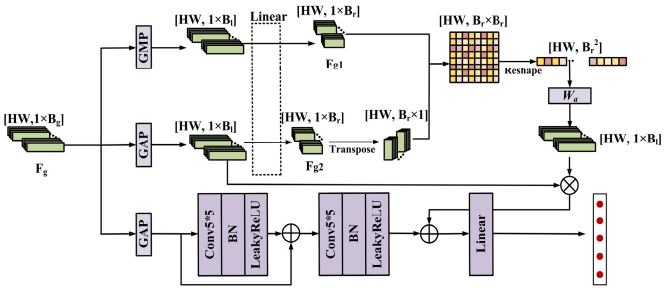


Fig. 4. Schematic of the proposed MERSA module. The operator $\otimes$ denotes element-wise multiplication. The operator $\oplus$ denotes the feature concatenation.

dimension is reduced to 64, a large matrix with dimensions of 377 856 $\times$ 64 $\times$ 64 would be required. Under conditions of limited computing resources, it is difficult to capture the inter-band correlations of HSI. Therefore, we design a MERSA module that adapts to the full-batch training manner to extract abundant spectral context and spatial features.

We first employ two consecutive pointwise convolution layers to reduce the spectral dimension. Then, the output $\mathbf{F}_g \in \mathbb{R}^{HW\times 1\times B_g}$ is fed into the MERSA module to extract abundant spectral–spatial features. As shown in Fig. 4, global average pooling (GAP) and global max pooling (GMP) are first conducted on $\mathbf{F}_g$ to further reduce the spectral dimension to B_l and extract channel-wise statistics of hyperspectral data. In the upper sub-branches of the MERSA module, the linear layer is used to reduce the spectral dimension and extract representative features $\mathbf{F}_{g1} \in \mathbb{R}^{HW\times 1\times B_r}$ and $\mathbf{F}_{g2} \in \mathbb{R}^{HW\times 1\times B_r}$. In our experiments, we set B_l to 64 and B_r to 8. It is worth noting that $B_r \ll B_l$, which is crucial for optimizing GPU memory usage. Then, the spectral attention map $\mathbf{M}_{\text{sam}} \in \mathbb{R}^{HW\times B_r\times B_r}$ is obtained as follows:

$$\mathbf{M}_{\text{sam}} = \mathbf{F}_{g1}\mathbf{F}_{g2}^{\text{T}}. \tag{17}$$

In order to apply the spectral attention map to the original spectral feature, we reshape $\mathbf{M}_{\text{sam}}$ into a vector feature and introduce a linear mapping matrix W_a to map the spectral dimension of this vector to B_l. This process can be expressed as

$$\mathbf{F}_s = \text{GAP}(\mathbf{F}_g) \bigotimes (W_a \cdot T(\mathbf{M}_{\text{sam}})) \tag{18}$$

where GAP$(\cdot)$ represents the global average pooling, $T(\cdot)$ is the reshape operation, and W_a is a linear projection matrix.

In the other sub-branch of the MERSA module, we utilize two convolutional layers with a kernel size of 5 $\times$ 5 to extract spatial features, employing padding operation to maintain consistency between the input and output spatial dimensions. Subsequently, the obtained contextual spectral features are concatenated with the spatial features to obtain the final output features. The output of MERSA can be directly concatenated with GCN features for final classification prediction.

E. Joint Loss

As shown in Fig. 2, the outputs of GCN and CNN branches are concatenated together, and then fed into the softmax classifier. After that, a cross-entropy loss term $\mathcal{L}_{\text{cls}}$ is used to perform the classification training, that is,

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N_t}\sum_{i=1}^{N_t}\sum_{c=1}^{C} y_{i,c}\log\left(p_{i,c}\right) \tag{19}$$

where C denotes the number of classes, N_t denotes the number of total training samples, $y_{i,c}$ represents the true label of the ith input samples, and $p_{i,c}$ represents the predicted output of the ith input samples.

Finally, we have the following joint loss to supervise the final classification score:

$$\mathcal{L}_{\text{joint}} = \eta_1\mathcal{L}_{r1} + \mathcal{L}_{\text{cls}} \tag{20}$$

where η_1 is learnable hyperparameter that balances the effect of the classification loss and the reconstruction loss.

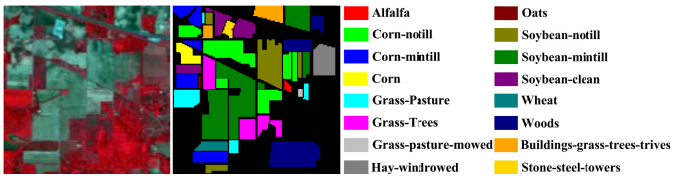


Fig. 5. Visualization of the Indian Pines dataset. (From left to right) False-color image, ground-truth map, and classes of land covers, respectively.

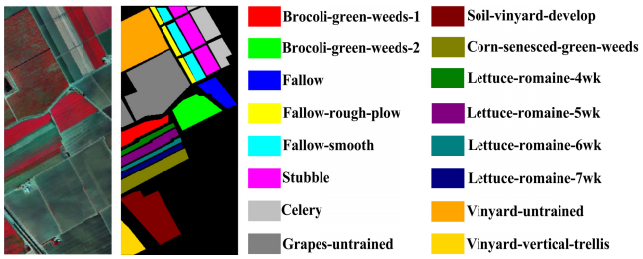


Fig. 6. Visualization of the Salinas dataset. (From left to right) False-color image, ground-truth map, and classes of land covers, respectively.

IV. Experiments

A. Experimental Datasets

We evaluate the proposed DSGCN on four different benchmark datasets. The first experiment is carried out on the Indian Pines dataset, an area of mixed agriculture and forestry in Northwestern Indiana. The spatial dimension comprises 145 $\times$ 145 pixels with a spatial resolution of 20 m. There are 200 bands in the spectral dimension with a spectral range of 0.4–2.5 μm after abandoning the noisy and water-absorbed channels. This dataset consists of 16 ground-truth classes, as shown in Fig. 5. The false-color image and ground-truth map are shown in Fig. 5(a) and (b), respectively.

The second dataset is captured by the AVIRIS sensor over Salinas Valley, California. The whole image contains 512 $\times$ 217 pixels and 204 spectral bands. The Salinas dataset consists of 16 ground-truth classes, as shown in Fig. 6. The false-color image and ground-truth map are shown in Fig. 6(a) and (b), respectively.

The third dataset is the WHU-Hi-HanChuan [24], acquired by a Leica Aibot X6 UAV V1 platform in HanChuan, Hubei, China. We intercepted a subregion with a size

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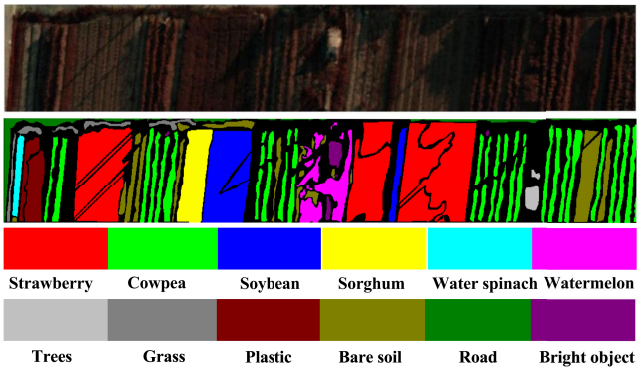


Fig. 7. Visualization of the WHU-Hi-HanChuan dataset. (From top to bottom) False-color image, ground-truth map, and classes of land covers, respectively.

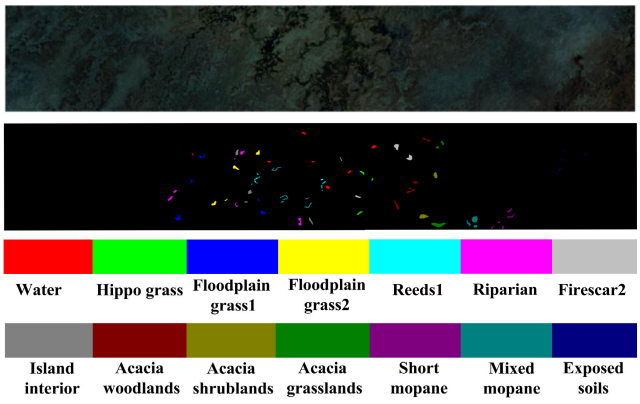


Fig. 8. Visualization of the Botswana dataset. (From top to bottom) False-color image, ground-truth map, and classes of land covers, respectively.

of 1100 × 150 pixels, which is a typical complex agricultural scene. There are 274 bands in the spectral dimension with a spectral range of 400–1000 nm. The intercepted WHU-Hi-HanChuan dataset consists of 12 classes, as shown in Fig. 7. The false-color image and ground-truth map are shown in Fig. 7(a) and (b), respectively.

The fourth dataset we use is the Botswana data, acquired by the NASA Earth Observing-1 satellite over the Okavango Delta, Botswana, on May 11, 2001. The original image contains 1476 × 256 pixels and 242 spectral bands. After removing the uncalibrated and noisy bands, we conduct the experiments using the remaining 145 bands. The Botswana dataset consists of 14 ground-truth classes, as shown in Fig. 6. The false-color image and ground-truth map are shown in Fig. 8(a) and (b), respectively.

For the four HSI datasets described above, we randomly select five labeled samples per class to evaluate the generalization capability of the proposed method.

B. Experimental Setup and Comparison Methods

Our experiments are implemented on the Pytorch (version 1.7.1) platform. The equipment used in our experiments consists of an Intel CPU i5-10400F, one NVIDIA GeForce GTX TITAN X (12 GB), and two 16-GB RAM. For model optimization, we use the Adam optimizer with a learning rate of 5e-4 and a weight decay of 0.02. The number of training epochs is set to 200. All experiments are carried out ten times independently to eliminate the influence of random sampling.

To evaluate the performance of the proposed DSGCN, especially in the case of few labeled samples, we choose several representative HSI classification methods for comparison, including SVM [3], SSRN [8], SF-SMF [44], DCFSL [23], S3Net [16], CEGCN [31], and WFCG [32]. We use the recommended hyperparameters in the original papers for fair comparison. The details of the compared methods are listed as follows.

1) SVM: A traditional machine learning-based method that maps data to a linearly separable high-dimensional feature space. However, this method ignores the 2-D spatial information of HSI.

2) SSRN: As a representative CNN-based method, SSRN can effectively extract spectral–spatial features from HSI cube data with the designed spectral and spatial residual blocks.

3) SF-SMF: SF-SMF is a transformer-based method. This method applies SpectralFormer [45] to encode spectrum motion features, efficiently improving classification performance. Due to the huge memory consumption of SF-SMF on large datasets, we only conducted experiments on the Indian Pines and Salinas datasets.

4) DCFSL: DCFSL is a classical transfer learning-based method. DCFSL solves the domain shift problem between the source and target datasets through an adversarial domain adaptive strategy. The Chikusei dataset is selected as the source domain dataset.

5) S3Net: S3Net is a few-shot classification method. S3Net adopts the idea of Siamese networks and achieves excellent classification performance in the case of a few labeled samples.

6) CEGCN, WFCG: CEGCN and WFCG belong to superpixel-based GCN methods. CEGCN is a hybrid deep learning (DL) framework that integrates CNN with GCN for pixel-level and superpixel-level feature fusion. Following the idea of CEGCN, WFCG proposes to combine GATs and CNN to explore critical features of HSI.

C. Comparison of Classification Performance

In this section, we display the classification results obtained by the proposed DSGCN and the comparative methods on the four datasets. We adopted overall accuracy (OA), average accuracy (AA), and Kappa as classification metrics for assessing the effectiveness of our method in the case of a few labeled samples. Tables II–V list the classification performance of all methods on the four datasets, where the best results for each metric are shown in bold. Their corresponding colorful classification maps are demonstrated in Figs. 9–12.

1) Results on the Indian Pines Dataset: As shown in Table II, it is difficult to infer satisfactory performance on the Indian Pines dataset under small sample conditions. For example, the classification accuracy of the SF-SMF algorithm is as low as 46.50% due to the scarcity of training samples for SpectralFormer. All DL-based algorithms perform better than SVM due to their powerful feature-extraction capabilities. The accuracies of DCFSL and S3Net are higher than SF-SMF and SSRN, which can be attributed to the positive impact of the training mechanisms of transfer learning and FSL. Compared

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TABLE II

OA, AA, INDIVIDUAL CLASS ACCURACY (%), AND KAPPA STATISTIC (KAPPA) OBTAINED BY DIFFERENT METHODS ON THE INDIAN PINES DATASET (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	SVM	SSRN	SF-SMF	DCFSL	S3Net	CEGCN	WFCG	DSGCN
1	16.76	63.61	24.32	**100.00**	97.80	93.08	94.78	97.69
2	32.21	**71.23**	33.98	57.27	52.23	44.96	65.05	60.95
3	25.90	63.33	35.75	58.42	44.76	43.35	69.29	**73.61**
4	19.17	40.56	40.99	77.59	86.85	69.26	93.29	**94.20**
5	61.91	**89.29**	52.90	83.89	83.18	78.16	79.71	78.51
6	83.60	94.45	71.21	93.66	91.96	97.99	**98.42**	95.14
7	22.45	17.75	13.68	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
8	95.54	93.42	83.34	88.16	85.52	87.82	**100.00**	98.51
9	8.52	40.03	10.96	**100.00**	98.00	98.83	**100.00**	**100.00**
10	40.21	45.16	41.66	72.29	64.99	63.97	**73.80**	73.12
11	51.26	69.02	62.04	37.47	59.49	63.08	73.33	**79.87**
12	26.82	50.76	24.95	61.56	63.76	46.37	75.09	**77.87**
13	86.14	73.45	52.20	98.00	90.80	99.75	99.80	**99.95**
14	92.80	**98.83**	83.91	95.16	86.22	89.50	91.15	87.25
15	21.77	55.16	60.65	59.32	80.08	64.11	91.35	**97.05**
16	79.63	71.89	20.98	97.73	99.77	99.67	98.06	**99.89**
OA	43.55	65.20	46.50	66.27	68.66	67.20	79.64	**80.48**
AA	47.79	64.87	44.60	80.03	80.34	77.50	87.70	**88.35**
Kappa×100	37.75	61.15	41.05	62.37	64.90	62.58	76.85	**77.83**

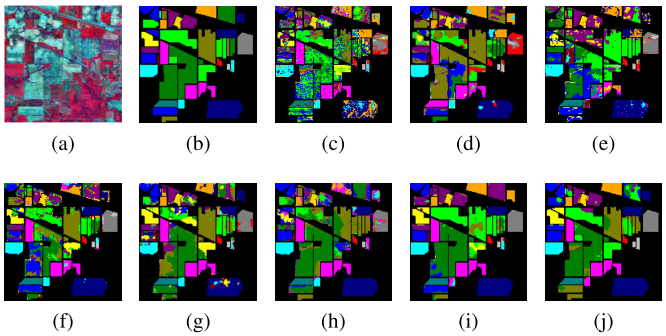


Fig. 9. Indian Pines dataset. (a) False color image. (b) Ground truth. (c)–(j) Classification maps produced using five labeled samples for each class. The methods used include: (c) SVM (OA = 43.55%); (d) SSRN (OA = 65.20%); (e) SF-SMF (OA = 46.50%); (f) DCFSL (OA = 66.27%); (g) S3Net (OA = 68.66%); (h) CEGCN (OA = 67.20%); (i) WFCG (OA = 79.64%); and (j) DSGCN (OA = 80.48%).

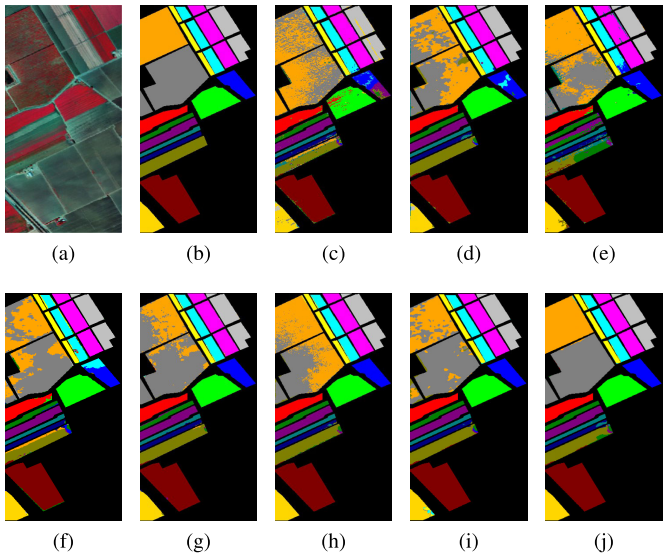


Fig. 10. Salinas dataset. (a) False color image. (b) Ground truth. (c)–(j) Classification maps produced using five labeled samples for each class. The methods used include: (c) SVM (OA = 75.56%); (d) SSRN (OA = 82.33%); (e) SF-SMF (OA = 82.52%); (f) DCFSL (OA = 88.50%); (g) S3Net (OA = 90.39%); (h) CEGCN (OA = 88.42%); (i) WFCG (OA = 92.52%); and (j) DSGCN (OA = 98.44%).

to the other algorithms, we can easily find that the proposed DSGCN obtains the highest OA of 80.48%. In particular, the classification accuracy of classes 1, 4, 6-9, 13, and 15-16 is more than 90%. In comparison with the GCN-based methods (e.g., CEGCN, WFCG), the OA of our method improved by 13.28% and 0.84%, respectively. To better display the classification performance of these methods, Fig. 9 visualizes the classification results. GCN models (e.g., CEGCN, WFCG, DSGCN) obtain relatively smooth classification maps, owing to the effect of superpixel segmentation. The result of DSGCN is more consistent with the ground truth, which further demonstrates the superiority of DSGCN. Moreover, the result of our method has the least noise.

2) Results on the Salinas Dataset: From Table III, the aforementioned methods achieve relatively good classification results on the sample-balanced dataset. The OA of DSGCN is 22.88%, 16.11%, 15.92%, 9.94%, 8.05%, 10.02%, and 5.92% higher than that of SVM, SSRN, SF-SMF, DCFSL, S3Net, CEGCN, and WFCG, respectively. DSGCN significantly improves the classification accuracies of class 8 and class 15 in comparison to the other methods, which demonstrates the effectiveness of DSGCN in dealing with land-cover classes containing some noisy pixels. It can be observed in Fig. 10 that the classification result of DSGCN exhibits less noise and more homogeneous regions. In comparison, DSGCN produces cleaner classification results for category 8 and category 10. In addition, we have the following observations for the GCN-related models. First, CEGCN, WFCG, and DSGCN obtain a much higher accuracy on the Salinas dataset than that on the Indian Pines dataset. One reason is that the high accuracy of superpixel segmentation on the Salinas dataset has a positive effect on the superpixel-based GCN models. Second, our proposed model DSGCN outperforms the GCN and GAT models on all three evaluation metrics, confirming that the joint use of spectral and SS information boosts the classification capability of GCN.

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TABLE III

OA, AA, INDIVIDUAL CLASS ACCURACY (%), AND KAPPA STATISTIC (KAPPA) OBTAINED BY DIFFERENT METHODS ON THE SALINAS DATASET (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	SVM	SSRN	SF-SMF	DCFSL	S3Net	CEGCN	WFCG	DSGCN
1	87.03	**100.00**	82.67	99.30	98.64	99.87	98.33	**100.00**
2	98.75	97.10	95.12	99.49	97.30	**100.00**	99.99	**100.00**
3	91.75	85.04	82.47	69.76	98.39	99.63	99.62	**100.00**
4	97.29	98.90	92.54	99.50	97.65	**99.81**	99.66	99.55
5	93.47	97.99	94.22	87.09	97.16	98.08	**98.54**	98.42
6	**100.00**	99.83	97.22	**100.00**	99.63	99.83	99.87	99.94
7	97.14	98.92	95.63	99.69	98.74	99.93	99.68	**100.00**
8	58.22	63.92	85.09	86.21	77.62	55.21	84.56	**98.35**
9	98.87	98.92	92.48	98.02	97.82	**100.00**	99.97	**100.00**
10	83.11	88.54	84.82	71.34	89.27	92.20	**95.67**	92.74
11	76.04	81.39	50.50	94.83	98.16	99.44	99.37	**99.77**
12	73.46	98.75	77.76	**100.00**	95.44	99.94	99.99	99.97
13	70.16	87.42	56.17	99.56	96.84	**99.95**	99.64	99.30
14	89.75	**99.46**	82.09	97.93	94.73	97.85	97.71	96.63
15	37.45	45.16	71.12	66.10	77.58	88.67	72.50	**95.57**
16	85.13	99.09	83.55	96.45	97.85	98.10	98.38	**99.93**
OA	75.56	82.33	82.52	88.50	90.39	88.42	92.52	**98.44**
AA	83.60	90.03	82.72	91.58	94.55	95.53	96.47	**98.76**
Kappa×100	72.94	80.36	80.65	87.19	89.33	87.19	91.66	**98.26**

TABLE IV

OA, AA, INDIVIDUAL CLASS ACCURACY (%), AND KAPPA STATISTIC (KAPPA) OBTAINED BY DIFFERENT METHODS ON THE WHU-HI-HANCHUAN DATASET (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	SVM	SSRN	SF-SMF	DCFSL	S3Net	CEGCN	WFCG	DSGCN
1	86.16	84.52	-	53.15	62.75	**88.29**	86.53	83.68
2	74.23	**88.88**	-	57.04	51.66	46.33	59.86	69.22
3	53.11	81.80	-	76.02	78.44	87.88	83.87	**93.45**
4	58.90	93.50	-	78.38	95.95	**98.41**	95.90	97.03
5	12.56	31.35	-	86.03	79.40	**97.01**	96.13	96.89
6	8.01	27.65	-	32.53	50.67	49.07	68.78	**91.57**
7	37.95	67.96	-	62.50	88.65	90.85	93.23	**93.92**
8	6.68	42.03	-	55.44	68.10	65.98	**81.59**	77.49
9	9.80	25.35	-	62.79	75.78	76.06	79.68	**94.23**
10	50.41	36.94	-	65.70	44.41	49.50	52.49	**70.65**
11	25.11	54.51	-	**75.69**	61.99	58.31	66.15	70.32
12	7.32	76.02	-	71.14	71.55	78.65	68.46	**79.69**
OA	38.69	57.61	-	60.25	62.67	71.88	75.56	**81.36**
AA	35.85	59.21	-	64.70	69.11	73.86	77.72	**84.84**
Kappa× 100	31.08	50.51	-	53.70	56.17	66.09	70.59	**77.54**

3) Results on the WHU-Hi-HanChuan Dataset: Table IV shows the detailed classification accuracies with different methods using five labeled samples per class. The complex distribution of ground objects and the phenomena of "same spectrum different object" within the WHU-Hi-HanChuan dataset bring more challenges. The DSGCN achieves the highest OA of 81.36% for the WHU-Hi-HanChuan dataset, with improvements of 9.48% and 5.80% over the CEGCN and WFCG methods, respectively. DSGCN reaches particularly high accuracy in some classes, such as soybean, sorghum, water spinach, watermelon, trees, and plastic. This also shows that the proposed method DSGCN can achieve promising results when facing the problems caused by the characteristics of the WHU-Hi-HanChuan dataset and the small number of labeled samples. Furthermore, we can see that many dot noises exist in the classification maps of SVM, SSRN, DCFSL, and S3Net in Fig. 11, while superpixel-based GCNs can acquire more reasonable maps than other comparison methods.

4) Results on the Botswana Dataset: Table V compares the experimental results of DSGCN to various baselines on the Botswana dataset. It can be seen that all algorithms, excluding SVM, achieve an OA above 90%, which demonstrates the importance of integrating spatial and spectral features. It is worth noting that DCFSL performs better on the Botswana dataset than on the other three datasets. A key reason is that the knowledge learned from the source dataset can be effectively transferred to the Botswana dataset. Leveraging no extra datasets, the proposed DSGCN exhibits better performance than DCFSL. In terms of OA, AA, and Kappa coefficients, the proposed DSGCN achieves the best results among all the evaluated methods. Fig. 12 shows the classification maps generated by different methods on the Botswana dataset. The classification map generated by DSGCN is noticeably closer to the ground-truth map than the other methods.

D. Ablation Studies

To evaluate the performance of major components of DSGCN, the ablation studies on the four datasets are conducted.

1) Effectiveness of Different Modules in the GCN Branch: Two modules in the GCN branch are critical for boosting the classification performance, i.e., the SS module and the SF module. In the SS module, the superpixel-level autoencoder and the KAN layer, which play crucial roles, are denoted as SLa and KAN, respectively. The classification accuracies with

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TABLE V

OA, AA, INDIVIDUAL CLASS ACCURACY (%), AND KAPPA STATISTIC (KAPPA) OBTAINED BY DIFFERENT METHODS ON THE BOTSWANA DATASET (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	SVM	SSRN	SF-SMF	DCFSL	S3Net	CEGCN	WFCG	DSGCN
1	**100.00**	96.51	-	**100.00**	96.30	99.96	98.63	99.92
2	82.46	89.09	-	**100.00**	98.75	99.69	99.21	**100.00**
3	80.46	91.42	-	**97.97**	92.52	97.79	92.35	97.71
4	62.96	96.21	-	**100.00**	99.52	99.81	99.86	99.67
5	82.28	91.89	-	**96.21**	79.89	84.30	87.25	93.45
6	62.87	80.57	-	87.88	81.33	85.61	**97.92**	93.96
7	96.25	**100.00**	-	**100.00**	94.88	93.67	98.38	99.49
8	93.49	99.70	-	**100.00**	99.65	97.91	**100.00**	98.01
9	79.62	92.32	-	99.68	93.69	**99.97**	99.24	98.04
10	62.03	97.69	-	97.94	93.62	99.26	91.81	**99.94**
11	**99.17**	98.03	-	95.00	92.50	92.30	92.66	96.39
12	90.53	94.88	-	98.86	98.81	99.89	99.78	**100.00**
13	89.27	89.94	-	99.62	99.09	99.13	**100.00**	**100.00**
14	95.00	**99.38**	-	93.33	98.33	93.55	97.79	96.75
OA	82.17	93.14	-	97.58	93.39	95.62	96.44	**97.96**
AA	84.03	94.11	-	97.61	94.21	95.92	96.78	**98.10**
Kappa× 100	80.71	92.57	-	97.37	92.84	95.21	96.05	**97.76**

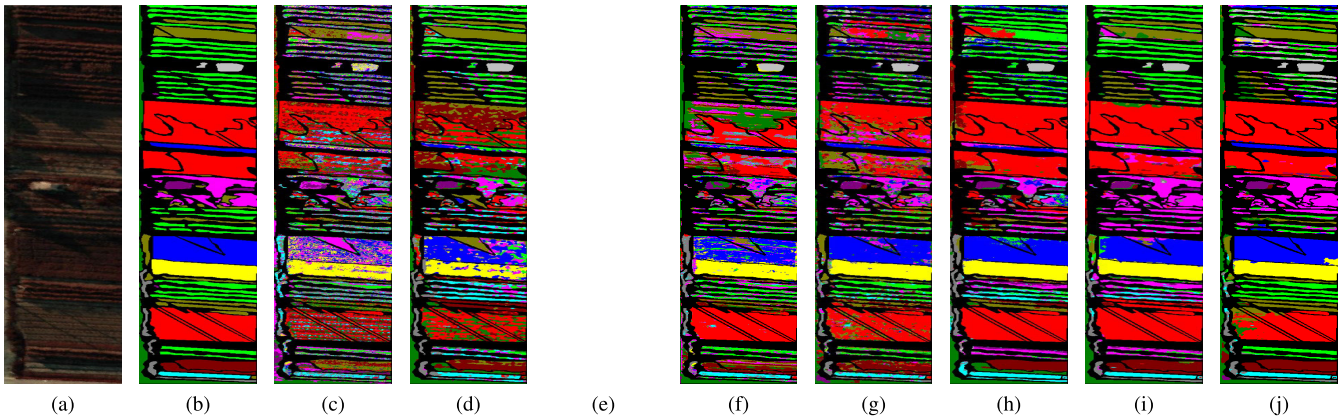


Fig. 11. Whu-Hi-HanChuan dataset. (a) False color image. (b) Ground truth. (c)–(j) Classification maps produced using five labeled samples for each class. The methods used include: (c) SVM (OA = 38.69%); (d) SSRN (OA = 57.61%); (e) SF-SMF; (f) DCFSL (OA = 60.25%); (g) S3Net (OA = 62.67%); (h) CEGCN (OA = 71.88%); (i) WFCG (OA = 75.56%); and (j) DSGCN (OA = 81.36%).

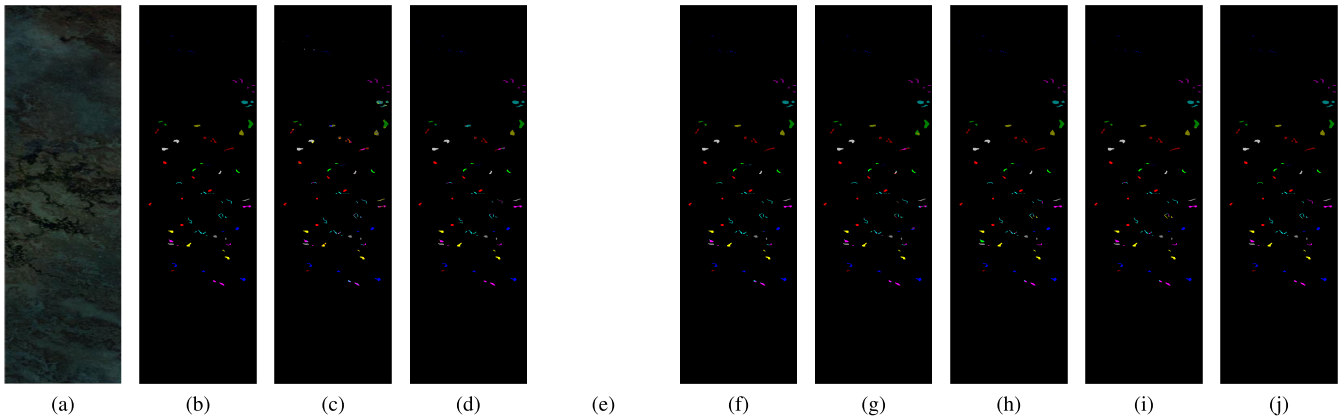


Fig. 12. Botswana dataset. (a) False color image. (b) Ground truth. (c)–(j) Classification maps produced using five labeled samples for each class. The methods used include: (c) SVM (OA = 82.17%); (d) SSRN (OA = 93.14%); (e) SF-SMF; (f) DCFSL (OA = 97.58%); (g) S3Net (OA = 93.39%); (h) CEGCN (OA = 95.62%); (i) WFCG (OA = 96.44%); and (j) DSGCN (OA = 97.96%).

the combination of SLa, KAN, and SF are shown in Table VI. When employing only the KAN layer, DSGCN achieves satisfactory results on the Salinas and Botswana datasets, with the OA being 8.02% and 1.76% higher than CEGCN, respectively. From the fourth row, we can observe similar conclusions regarding the Botswana dataset. This indicates that the KAN layer can enhance the nonlinear expression capability of GCN. Combining the KAN layer with either the SLa or the SF module significantly improves the classification performance on the first three datasets compared to using only the KAN layer. Particularly, the introduction of the SF module significantly enhances the classification accuracy of

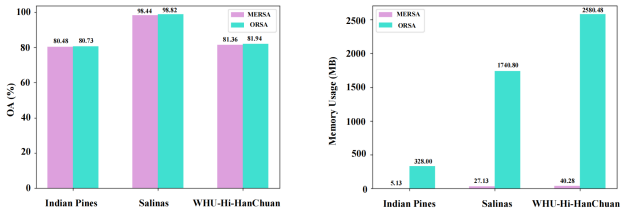
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TABLE VI
OA (%) INDICES OF THE COMBINATIONS OF DIFFERENT MODULES

KAN	SLa	SF	Indian Pines	Salinas	Whu-Hi-HanChuan	Botswana
✓	✘	✘	73.22	96.44	76.38	97.38
✓	✓	✘	74.36	97.33	77.68	97.53
✓	✘	✓	79.02	98.43	79.89	97.34
✘	✓	✓	79.63	96.94	80.18	96.82
✓	✓	✓	**80.48**	**98.44**	**81.36**	**97.96**

TABLE VII
OA (%) INDICES OF DSGCN WITH AND WITHOUT THE MERSA MODULE

Dataset	Without MERSA	With MERSA
Indian Pines	69.61	**80.48**
Salinas	98.27	**98.44**
Whu-Hi-HanChuan	78.50	**81.36**
Botswana	93.70	**97.96**



(a) (b)

Fig. 13. Performance comparison between MERSA and ORSA modules on three datasets. (a) OA (%) indices. (b) Memory usage (MB) when computing the attention map.

the model on the Indian Pines dataset. The primary reason is that the SF module offers novel perspectives on spectra recognition, which is advantageous for tackling classification challenges in imbalanced datasets. In addition, integrating the three modules yields the highest classification outcomes across the four datasets, demonstrating their complementary nature.

2) Effect of the MERSA Module: We perform a series of experiments to verify the effectiveness of the MERSA module. As shown in Table VII, with the incorporation of the MERSA module, our network achieves significant performance improvements on the other three datasets apart from the Salinas dataset. The main factor is that the GCN branch already achieves adequate performance on the Salinas dataset, making the incorporation of the MERSA module relatively inconsequential to the classification result. Furthermore, we compare the classification performance and GPU memory usage between the MERSA module and the attention module using the original spectral dimension (called the ORSA module). As shown in Fig. 13, compared to the ORSA module, the use of our proposed MERSA module results in a slight decrease in classification accuracies across the three datasets, while significantly reducing GPU memory usage and enhancing the algorithm's applicability. Additionally, we find that as the spatial dimensions of a dataset increase, the advantage of our method in terms of GPU memory usage becomes more significant.

E. Impact of the Number of Training Samples

We evaluate the classification accuracies of the proposed DSGCN and the compared methods when using different numbers of labeled samples for training in this experiment. As shown in Fig. 14, the number of labeled samples per category is set from 5 to 25 with an interval of 5. We observe that the classification performance of all the methods can be generally improved when the number of training samples increases. Our proposed DSGCN produces pleasing results compared with the other competing methods. Especially, our model stands out for its stability on all four datasets, even as the number of samples changes. DSGCN achieves similar results on the Indian Pines dataset compared to the most advanced algorithm, WFCG. In addition, it is noteworthy that DSGCN has more obvious advantages than the other methods on the Salinas and WHU-Hi-HanChuan datasets. This is mainly due to the utilization of unlabeled pixels during graph convolution, as well as the extraction of spectral and SS information. Given ten labeled samples per class on the Botswana dataset, DCFSL demonstrates a more competitive performance compared to DSGCN, indicating that DCFSL is more sensitive to the quantity of labeled samples. Based on the above observations, we can conclude that DSGCN effectively constrains the overfitting problem under scenarios with few labeled samples.

F. Influence of Segmentation Scale

The superpixel segmentation scale λ is an important hyperparameter that controls the average size of superpixels and determines the quality of node features. In other words, a smaller value of λ results in a higher number of superpixels, yielding a more refined segmentation map. We evaluate the performance of DSGCN at different segmentation scales in this experiment. Due to the differences in spatial resolution of the four datasets, we choose different scale criteria for each dataset to conduct this experiment. We set the segmentation scale to 100, 200, 300, 400, and 500 for the Indian Pines dataset, 100, 500, 900, 1300, and 1700 for the Salinas dataset, 100, 500, 900, 1300, and 1700 for the WHU-Hi-HanChuan dataset, and 500, 900, 1300, 1700, and 2100 for the Botswana dataset, respectively. When λ varies, the proposed DSGCN achieves stable performance on the four datasets. Theoretically, as λ increases, the superpixel segmentation map becomes coarser, and the classification performance deteriorates. As shown in Fig. 15, a higher value of λ on the Indian Pines dataset corresponds to a lower OA. However, the OA is slightly improved on the other three datasets when λ increases. The main reason is that the over-correction of our method depresses the classification accuracy when obtaining an accurate superpixel segmentation map. Additionally, if λ is too large, each node in the graph may contain several categories, decreasing the classification performance. Therefore, we can conclude that an appropriate scale is more beneficial to the classification results. We fix the segmentation scale λ to 100 on the Indian Pines dataset, 500 on the Salinas and WHU-Hi-HanChuan datasets, and 900 on the Botswana dataset for the remaining experiments.

G. Evaluation of the Roles of Different Subnetworks in DSGCN

In this experiment, we conduct the ablation experiment to verify the effectiveness of different subnetworks. Table VIII

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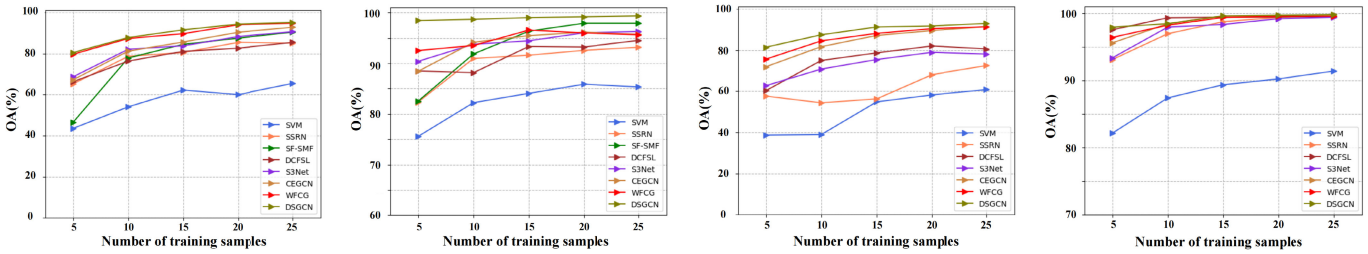


Fig. 14. OA (%) indices of various algorithms in case of different labeled sample numbers per class. (a) Indian Pines dataset. (b) Salinas dataset. (c) WHU-Hi-HanChuan dataset. (d) Botswana dataset.

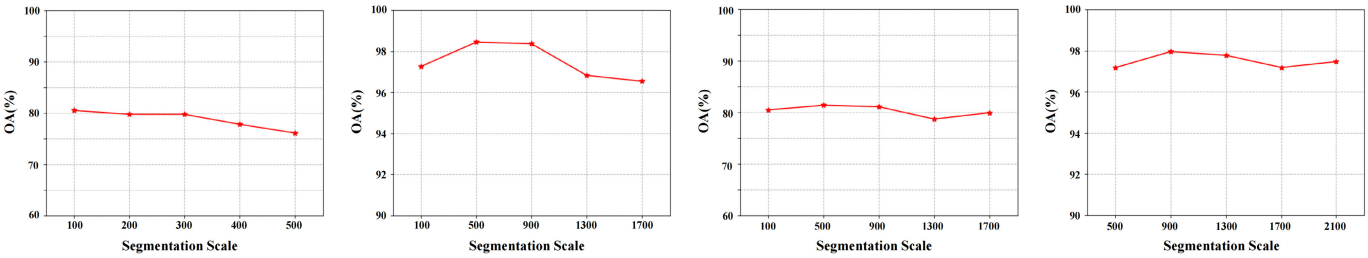


Fig. 15. OA (%) indices of DSGCN with different segmentation scales on each dataset. (a) Indian Pines dataset. (b) Salinas dataset. (c) WHU-Hi-HanChuan dataset. (d) Botswana dataset.

TABLE VIII
OA (%) INDICES OF DIFFERENT SUBNETWORKS

Dataset	CNN	GCN	DSGCN
Indian Pines	72.25	72.37	**80.48**
Salinas	90.58	95.72	**98.44**
Whu-Hi-HanChuan	67.39	71.94	**81.36**
Botswana	96.42	85.25	**97.96**

shows that the CNN subnetwork and GCN subnetwork obtain similar performance on the Indian Pines dataset. For the Salinas and WHU-Hi-HanChuan datasets, the GCN subnetwork achieves better performance, which reveals that GCN can effectively implement the HSI classification by making full use of the spatial information through the superpixel segmentation algorithm. For the Botswana dataset, the GCN subnetwork significantly underperforms compared to the CNN subnetwork. As shown in Fig. 12, the distribution of ground objects in the Botswana dataset is highly dispersed, making it challenging for superpixel-based GCN to establish effective connectivity between nodes. We can further conclude that the concatenation of GCN and CNN features has a positive contribution to the classification task.

H. Robustness Analysis

To comprehensively evaluate the robustness and generalization ability of different methods, we verify the impact of two types of noise (sample noise and label noise) on model performance. The experiments are conducted on the Indian Pines dataset.

1) Analysis of Robustness Under Sample Noise: During the testing phase, we add Gaussian noise in the spectral dimension to simulate different SNR conditions. To ensure a fair comparison and verify the SLa module's role in enhancing the robustness of GCN, we remove the SF module in this experiment. Table IX shows the AA at various SNRs for each model. For the low SNR interval of -5 to 5 dB, DCFSL achieves the highest accuracy due to the introduction of source knowledge and data. The results of GCN methods (CEGCN and WFCG) are not good. These GCN methods are trained in a full-batch manner (with the batch size equal to the entire dataset). If noisy datasets are used during the testing process, the graph structure will be disrupted, and the association information between nodes will become meaningless, leading to a sharp decline in performance. Compared to CEGCN and WFCG, our method shows significant advantages in robustness and generalization, indicating that the proposed SLa module plays an important role in enhancing the robustness of GCN.

2) Analysis of Robustness under Label Noise: In this experiment, we selected 25 training samples from each class within the Indian Pines dataset for comparative analysis. To simulate the noisy scenarios, a fixed number of samples (4, 8, and 12) from each class are selected and uniformly flipped to any other classes [46]. The experimental results are reported in Table X. As the number of noisy samples increases, a discernible degradation in performance can be observed across all models. Notably, the S3net exhibited the most severe degradation, primarily attributed to its reliance on sample pairs as input. With the escalation of noise, the pairing operation is likely to generate an increased number of noisy pairs. From Table X, it can also be observed that, compared to the other methods, DSGCN achieves satisfactory classification results under different amounts of noisy samples.

I. Computational Complexity

As shown in Table XI, the training and testing time, as well as the floating-point operations (FLOPs) and the number of

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TABLE IX
OA (%) INDICES OF DIFFERENT METHODS WITH VARIOUS SNR CONDITIONS ON THE INDIAN PINES DATASET

	SVM	SSRN	SF-SMF	DCFSL	S3Net	CEGCN	WFCG	DSGCN
-5 dB	2.28	11.79	5.23	**36.66**	12.96	6.64	4.58	23.01
0 dB	6.89	23.43	9.58	**56.36**	13.19	9.02	4.89	39.62
5 dB	18.21	45.63	10.92	**65.59**	13.40	7.53	5.94	64.22
10 dB	45.41	61.82	7.73	67.00	12.11	16.10	5.50	**73.21**

TABLE X
OA (%) INDICES OF DSGCN WITH DIFFERENT NUMBER OF MISLABELED SAMPLES ON THE INDIAN PINES DATASET

The number of noisy training samples	SVM	SSRN	SF-SMF	DCFSL	S3Net	CEGCN	WFCG	DSGCN
25(clean)+0(noisy)	65.49	85.07	90.43	85.46	90.63	92.82	94.79	**95.12**
21(clean)+4(noisy)	62.42	69.71	77.34	60.28	70.98	85.96	86.43	**87.42**
17(clean)+8(noisy)	56.54	64.05	66.68	44.09	49.78	**79.52**	76.40	79.01
13(clean)+12(noisy)	53.02	54.59	51.19	31.59	29.37	70.94	73.12	**74.51**

TABLE XI
COMPUTATIONAL TIME, FLOPS, AND PARAMETERS ON EACH DATASET WITH DIFFERENT METHODS

Dataset		SVM	SSRN	SF-SMF	DCFSL	S3Net	CEGCN	WFCG	DSGCN
Indian Pines	Training time (s)	2.44	29.57	326.97	226.81	41.12	17.71	27.12	94.22
	Testing time (s)	0.78	8.72	44.32	0.35	8.65	0.41	0.41	0.79
	FLOPs (G)	-	1.53	0.75	0.67	1.30	1.62	1.62	1.78
	#params (K)	-	364.17	2298.88	50.50	154.98	166.37	76.27	173.57
Salinas	Training time (s)	2.41	28.27	345.54	241.78	41.55	105.70	223.29	488.00
	Testing time (s)	4.03	47.33	240.66	1.61	46.70	2.11	2.40	4.15
	FLOPs (G)	-	1.56	0.77	1.36	0.64	8.58	8.62	9.36
	#params (K)	-	370.31	2387.97	50.50	154.98	166.89	76.78	174.09
Whu-Hi-HanChuan	Training time (s)	1.98	28.34	-	224.99	27.63	255.76	724.40	536.40
	Testing time (s)	7.29	122.07	-	3.75	83.17	3.32	3.56	5.55
	FLOPs (G)	-	2.11	-	0.50	0.34	14.19	14.28	15.35
	#params (K)	-	477.73	-	49.85	92.37	175.47	85.62	182.67
Botswana	Training time (s)	1.70	21.33	-	196.48	35.63	337.59	1316.10	1141.35
	Testing time (s)	0.22	2.06	-	0.09	13.422	6.46	6.83	11.13
	FLOPs (G)	-	1.11	-	0.58	0.34	26.18	26.31	28.76
	#params (K)	-	281.17	-	50.17	92.95	158.96	68.98	166.16

parameters, are analyzed to evaluate the computational complexity of different methods. We can observe from Table XI that SF-SMF exhibits the highest parameter count, and its training time is longer than the other methods, mainly due to self-attention and multihead attention operations. Compared with other methods, we can also observe that the GCN-based methods (e.g., CEGCN, WFCG, DSGCN) have more FLOPs because of the full-batch training manner. Moreover, the training time for the GCN-based methods increases with the expansion of the spatial dimensions of the datasets. Despite having a relatively long training time and more FLOPs in most cases, DSGCN also has the best classification performance.

J. Visualization

To observe the distribution of class features obtained by DL-based methods, we adopt the t-distributed stochastic neighbor embedding (t-SNE) [47] for visualization on the Salinas dataset. We randomly select 300 pixels from each class in the test dataset for clear observation. As shown in Fig. 16, it can be observed that the feature vectors from class 8 exhibit more overlap with class 15 for other methods, while our proposed DSGCN effectively addresses this issue. In addition, we can also observe that the feature vectors of the other methods exhibit higher intra-class concentration than the GCN-based methods.

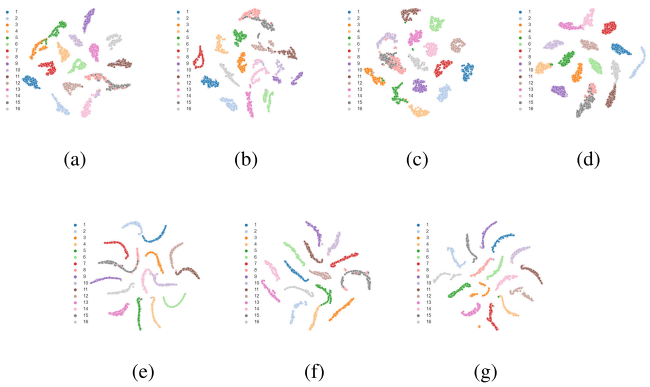


Fig. 16. Visualization of class features of different methods on the Salinas dataset. (a) SSRN. (b) SF-SMF. (c) DCFSL. (d) S3Net. (e) CEGCN. (f) WFCG. (g) DSGCN.

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V. Conclusion

In this article, we propose a novel DSGCN for HSI classification when using a small amount of labeled data. In the GCN branch, we design the SS module and SF module to explore the high-level semantic information of HSI. In the SS module, we integrate a superpixel-level reconstruction component into GCN, where the edge weights of the graph can be updated when training. In the SF module, the global spectral variation information of HSI is introduced to improve the classification accuracy. In this way, the nodes of the graph are represented by high-level semantic features of the image regions. In the CNN branch, a new global contextual feature extractor, MERSA, is designed to concatenate the GCN subnetwork for feature fusion and classification. Experimental results on four datasets show that our method achieves satisfactory performance compared with the other advanced approaches, especially when the number of training samples is rather limited.

The notable deficiency in the robustness of superpixel-based GCN methods indicates that enhancing their robustness under low SNR conditions will be a key focus in our future work.

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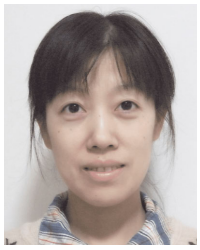
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